# AI Video on Grace Hopper

*Algorithm and Description*

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## 1 About the Scientist

Grace Murray Hopper was an American mathematician, computer scientist and officer of the US Navy. She is remembered as one of the first people who tried to make programming easier for ordinary people. In her time, programmers had to write long lists of numbers (machine code), and one small mistake could spoil the whole program. Grace Hopper felt that computers should understand words closer to human language, and she worked on this idea for many years.

| | |
|---|---|
| **Born** | 9 December 1906, New York, USA |
| **Died** | 1 January 1992 |
| **Education** | PhD in Mathematics, Yale University (1934) |
| **Career** | US Navy Reserve (from 1943), later retired as Rear Admiral |
| **Known for** | First compilers (A-0), FLOW-MATIC, and her role in the creation of COBOL |
| **Nickname** | “Amazing Grace” |

## 2 Her Main Contributions

- **Early computers.** She worked on the Harvard Mark I and later on the UNIVAC, which were among the first large computers.
- **Compiler.** In 1952 she developed the A-0 system, an early compiler that translated instructions into machine code automatically.
- **FLOW-MATIC.** This was one of the first languages that used English-like words for commands.
- **COBOL.** Her ideas influenced COBOL (1959), a language made for business data such as records, payroll and accounts.
- **Debugging.** She helped make the word “debugging” popular after a real moth was found stuck inside a computer relay in 1947.



## 3 Video Idea

Instead of a normal speech, the video is made as a conversation between Grace Hopper and a young student in a computer museum. The student asks simple questions (why was programming hard, what are punched cards, what is a compiler, what is COBOL), and Grace answers them. This way the topic is explained step by step and is easy for a viewer to follow. The video is 13 scenes $\times$ 5 seconds = 65 seconds long.

## 4 Tools Used

| Tool | Used for | Output |
|---|---|---|
| **DigEn AI** | Generating each 5 second clip from a text prompt and the reference image | 13 video clips |
| **CapCut** | Joining the clips in order, adding transitions and music | One 65 second video |
| **LaTeX (TikZ)** | Drawing the flowchart and preparing this PDF | Flowchart and report |

## 5 Algorithm of Video Making

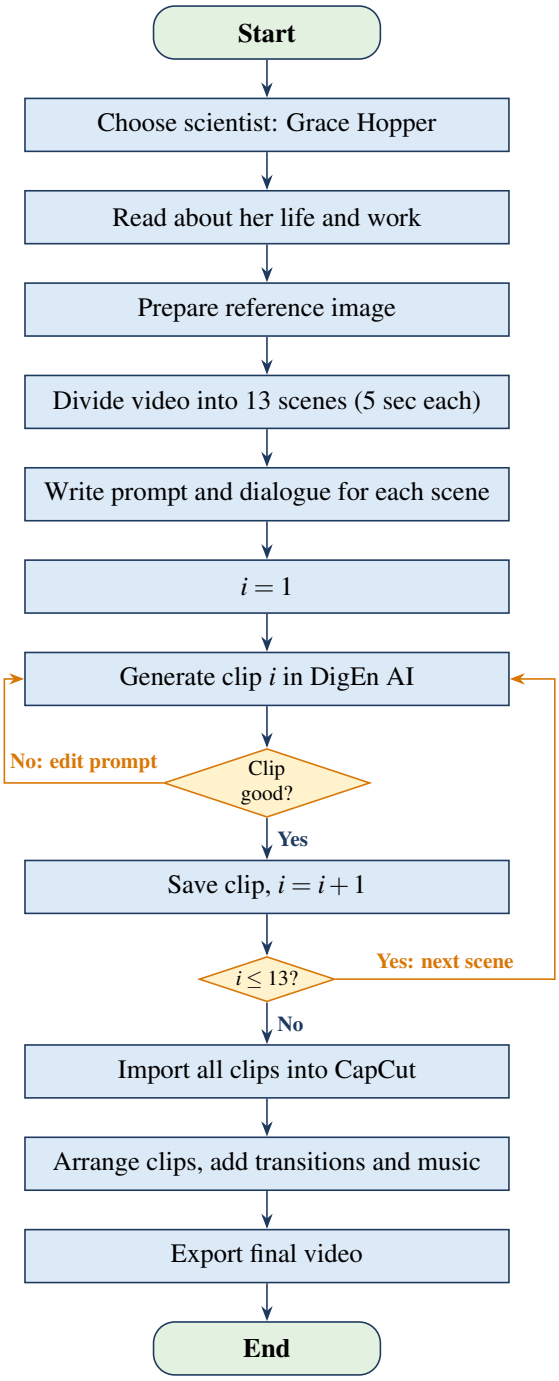
1. Start.
2. Choose the scientist: Grace Hopper.
3. Read about her life and her work on compilers and COBOL.
4. Take a reference image with Grace and the student, so that faces and dress stay the same in every scene.
5. Divide the video into 13 scenes of 5 seconds each.
6. Write a prompt and one short dialogue for each scene (a dialogue of about 12–14 words fits in 5 seconds).
7. Set $i = 1$.
8. Give the prompt of scene $i$ to DigEn AI and generate the clip.
9. If the clip is not good (face or dress changed, lip movement wrong), edit the prompt and generate again.
10. Save the clip and increase $i$ by 1.
11. If $i \leq 13$, go back to step 8.
12. Import all 13 clips into CapCut and arrange them in order.
13. Add transitions and background music.
14. Export the final video.
15. End.

## 6 Algorithm of Grace Hopper's Compiler (A-0 style)

The video is about compilers, so this is a simple version of how an early compiler works. In the A-0 system the programmer wrote subroutine numbers with their values, and the compiler joined the machine code of each subroutine from a library into one program.

```
START
  read the program (list of subroutine numbers with values)
  output <- empty
  FOR each instruction in the program DO
      code <- find subroutine (instruction.number) in library
      put instruction.values into code
      add code to output
  END FOR
  save output as machine code
END
```

## 7 Flowchart


Start
Choose scientist: Grace Hopper
Read about her life and work
Prepare reference image
Divide video into 13 scenes (5 sec each)
Write prompt and dialogue for each scene
$i = 1$
Generate clip $i$ in DigEn AI
Clip good?
No: edit prompt
Yes
Save clip, $i = i + 1$
$i \leq 13$?
Yes: next scene
No
Import all clips into CapCut
Arrange clips, add transitions and music
Export final video
End

## 8 Scenes and Dialogues

Grace stands on the left and the student on the right in all scenes. Neither looks at the camera, and only the named speaker talks.

| Scene | Speaker | Dialogue (5 seconds) |
|---|---|---|
| 1 | **Grace** | “Hello! I’m Grace Hopper.” |
| 2 | **Student** | “Grace, why was programming so difficult in the early days?” |
| 3 | **Grace** | “Early computers understood only machine code, so instructions were long and complicated.” |
| 4 | **Student** | “Did programmers really use punched cards to give instructions?” |
| 5 | **Grace** | “Yes! Each card’s holes represented instructions the computer could read.” |
| 6 | **Student** | “Was there an easier way to program computers?” |
| 7 | **Grace** | “A compiler translates human-friendly language into machine code.” |
| 8 | **Grace** | “My work helped advance compilers and influenced COBOL.” |
| 9 | **Student** | “Grace, what is COBOL, and why did you help develop it?” |
| 10 | **Grace** | “COBOL uses English-like commands to process business data easily.” |
| 11 | **Student** | “Now I understand! COBOL made programming easier to read and manage.” |
| 12 | **Grace** | “Exactly. Programming should be easy for everyone to use.” |
| 13 | **Student** | “Thank you, Grace! This was truly inspiring.” |

### Common part of every prompt

Use the same reference image to preserve both characters’ exact faces, ages and outfits. Grace Hopper stands on the left wearing her dark US Navy uniform, glasses and officer cap. The student stands on the right wearing black glasses, a navy-blue hoodie and a gray T-shirt. They stand face-to-face in a historical computer museum in a wide 16:9 frame. Natural body movement and lip-sync. Neither character looks at the camera. No subtitles, no text, no change in face or outfit.